Hints: Number theory – 6

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. The first two sums are multiplicative functions of n, so it suffices to check prime powers. $\sum_{d\mid p^k}\mu(d)^2 = 2$ and $\sum_{d\mid p^k}\mu(d)/d = 1-\frac{1}{p}$. Multiplying over $p \mid n$ gives $2^{\omega(n)}$ and $\varphi(n)/n$.

Hint 2. For the logarithm, write $\ln d = \ln n - \ln(n/d)$. Then $\sum_{d\mid n}\mu(d)\ln d = \ln n\cdot[n=1] - (\mu * \ln)(n)$. Show $\ln = \Lambda * 1$ by factoring n; inversion then gives $\Lambda = \mu * \ln$, so the sum is $-\Lambda(n)$.

Hint 3. Dirichlet series: $\sum \ln n\, n^{-s} = -\zeta'(s)$, and $\Lambda = \ln * \mu$ gives $\sum \Lambda(n)n^{-s} = -\zeta'(s)/\zeta(s)$.

Problem 2

Hint 1. Let N_d be the number of monic irreducible polynomials of degree d over $\mathbb{F}_q$. Show that $x^{q^n}-x$ is the product of all monic irreducibles whose degree divides n (each appears once, since the derivative is -1).

Hint 2. Comparing degrees gives $q^n = \sum_{d\mid n} dN_d$. Möbius inversion yields $nN_n = \sum_{d\mid n}\mu(d)q^{n/d}$.

Hint 3. Positivity: $nN_n \geqslant q^n - \sum_{j\leqslant n/2} q^j > q^n - q^{\lfloor n/2\rfloor+1} \geqslant 0$ for $n \geqslant 2$; the case $n=1$ is trivial.

Problem 3

Hint 1. Burnside: rotation by j fixes exactly the words that are constant on the cycles of $i \mapsto i+j$. There are $\gcd(j,n)$ such cycles, so it fixes $k^{\gcd(j,n)}$ words. The number of j with $\gcd(j,n) = n/d$ is $\varphi(d)$, so the count is $\frac{1}{n}\sum_{d\mid n}\varphi(d)k^{n/d}$.

Hint 2. Möbius: let $A(e)$ be the number of words of length e with least period exactly e (aperiodic words). Every word of length n is a repetition of a unique aperiodic word of length $e \mid n$, so $k^n = \sum_{e\mid n} A(e)$ and $A = \mu * k^{(\cdot)}$.

Hint 3. A necklace of length n with primitive period e is an orbit of size e, so the number of necklaces is $\sum_{e\mid n} A(e)/e$. Substitute $A(e) = \sum_{c\mid e}\mu(e/c)k^c$, collect the coefficient of k^c, and use $\sum_{m\mid n/c}\mu(m)\frac{n/c}{m} = \varphi(n/c)$.

Problem 4

Hint 1. $\frac{d(n)}{n^\varepsilon} = \prod_{p^a\parallel n}\frac{a+1}{p^{a\varepsilon}}$. For $p \geqslant 2^{1/\varepsilon}$ each factor is $\leqslant 1$. For the finitely many smaller p, each factor is bounded by a constant depending on ε.

Hint 2. For the lim sup, the lower bound uses primorials. Take $n = p_1\cdots p_k$; then $d(n) = 2^k$ and $\ln n = \theta(p_k) \sim k\ln k$. Hence $\ln d(n)\cdot\ln\ln n/\ln n \to \ln 2$.

Hint 3. Upper bound: run the argument of Hint 1 with $\varepsilon = (1+\delta)\ln 2/\ln\ln n$. The primes $p < 2^{1/\varepsilon}$ number at most $2^{1/\varepsilon} = (\ln n)^{1/(1+\delta)}$. Each contributes at most a factor $O(\ln n)$, so their total is $\exp(o(\ln n/\ln\ln n))$.

Hint 4. $\frac{\sigma(n)}{n} < \prod_{p\mid n}\frac{p}{p-1}$, and the product is largest when the primes of n are the smallest ones. Mertens' theorem $\prod_{p\leqslant y}\frac{p}{p-1} \sim e^\gamma \ln y$ then gives the maximal order $\limsup \frac{\sigma(n)}{n\ln\ln n} = e^\gamma$ (Gronwall); test it on $n = \operatorname{lcm}(1,\dots,k)$. Elementarily you get $\sigma(n)/n \asymp \ln\ln n$ along such n.

Problem 5

Hint 1. Sum $\mu * 1 = \varepsilon$ over $n \leqslant x$: $\sum_{d\leqslant x} \mu(d)\lfloor x/d\rfloor = 1$. Replace $\lfloor x/d\rfloor$ by $x/d - \{x/d\}$. This gives $\left|x\sum_{d\leqslant x}\mu(d)/d\right| \leqslant 1 + (x-1)$, so the sum is at most 1 in absolute value.

Hint 2. $\sum_{n\geqslant 1}\mu(n)/n = 0$ is equivalent to the prime number theorem (Landau); you only need to state this.

Hint 3. $\sum_{n\leqslant x}\lfloor x/n\rfloor$ counts pairs (n,k) with $nk \leqslant x$, and so does $\sum_{m\leqslant x} d(m)$. Using $\lfloor x/n\rfloor = x/n + O(1)$ gives $x\ln x + O(x)$. Splitting the pairs at $\sqrt{x}$ (the hyperbola method) recovers the term $(2\gamma - 1)x$.

Problem 6

Note. As printed the first claim is false: $f(n) = 1/n$ is (completely) multiplicative, tends to 0, and never vanishes. The hints prove the correct statement: for multiplicative f, $f(n) \to 0$ if and only if $f(p^k) \to 0$ as the prime power $p^k \to \infty$. Also, a completely multiplicative f with $|f| = 1$ is just an arbitrary choice of unimodular values at the primes.

Hint 1. One direction is trivial, since prime powers are integers. For the converse, fix $\eta \in (0,1)$. Only finitely many prime powers have $|f(p^k)| \geqslant \eta$, and these are bounded by some $M \geqslant 1$.

Hint 2. Write $n = \prod p^k$. If all prime powers in n are $\leqslant Y$, then $n \leqslant Y^{\pi(Y)}$. So every large n has a prime-power factor $> Y$, and then $|f(n)| \leqslant M^C \cdot \sup_{q>Y}|f(q)|$, where C is the number of exceptional prime powers. Let $Y \to \infty$.

Hint 3. Completely multiplicative functions are determined by their values at primes, which can be prescribed freely. So the f with $|f| = 1$ are exactly $f(n) = \prod p^k \mapsto \prod \zeta_p^k$ with arbitrary $|\zeta_p| = 1$; for instance n^{it}, or Liouville's $(-1)^{\Omega(n)}$.

Problem 7

Hint 1. Check that $f(x) = ax^m$ ($a, m \geqslant 0$ integers) works. Now suppose $f = x^m g$ with $g(0) \neq 0$ and $\deg g \geqslant 1$. Infinitely many primes divide some value $g(k)$ (Schur), so pick such a p with $p \nmid f(1)\,g(0)$; then $p \nmid k$.

Hint 2. Iterate the condition. Since $\mathrm{rad}(n^{\mathrm{rad}(n)}) = \mathrm{rad}(n)$, you get $\mathrm{rad}(f(n)) \mid \mathrm{rad}(f(n^{\mathrm{rad}(n)^j}))$ for every j.

Hint 3. By CRT choose $n \equiv k \pmod p$ with $\mathrm{rad}(p-1) \mid n$. Then $p \mid f(n)$, and for large j, $(p-1) \mid \mathrm{rad}(n)^j$, so $n^{\mathrm{rad}(n)^j} \equiv 1 \pmod p$. Hence $p \mid f(1)$, a contradiction. Answer: $f(x) = ax^m$.

Problem 8

Hint 1. $\frac{\varphi(n)}{n} = \sum_{d\mid n}\frac{\mu(d)}{d}$. So $\sum_{n\leqslant x}\frac{\varphi(n)}{n} = \sum_{d\leqslant x}\frac{\mu(d)}{d}\lfloor x/d\rfloor = x\sum_{d\leqslant x}\frac{\mu(d)}{d^2} + O(\sum_{d\leqslant x}\frac{1}{d})$. The tail of $\sum \mu(d)/d^2$ is $O(1/x)$, and the full sum is $6/\pi^2$.

Hint 2. Farey fractions of order N in $[0,1]$ are counted by $1 + \sum_{n\leqslant N}\varphi(n)$. The same Möbius computation gives $\sum_{n\leqslant N}\varphi(n) = \sum_d \mu(d)\frac{\lfloor N/d\rfloor(\lfloor N/d\rfloor+1)}{2} = \frac{3}{\pi^2}N^2 + O(N\ln N)$.

Hint 3. Relation: either sum follows from the other by partial (Abel) summation, since $\varphi(n) = n\cdot\frac{\varphi(n)}{n}$. The error $O(\ln x)$ becomes $O(N\ln N)$.

Problem 9

Hint 1. By multiplicativity, $\frac{\sigma(n)\varphi(n)}{n^2} = \prod_{p^a\|n}\left(1 - \frac{1}{p^{a+1}}\right)$.

Hint 2. Each factor lies in $[1-p^{-2}, 1)$, and $\prod_p (1-p^{-2}) = 1/\zeta(2) = 6/\pi^2$. Hence $\frac{6}{\pi^2} < \frac{\sigma(n)\varphi(n)}{n^2} < 1$ for $n > 1$; the left inequality is strict because the primes not dividing n contribute factors < 1.

Hint 3. The ratio $n^2/(\sigma\varphi)$ is close to 1 when all prime factors of n are large (e.g. n a large prime). It is close to $\pi^2/6$ when n is divisible by high powers of all small primes, e.g. $n = (k!)^k$ with k large. For example, $n = 2$ gives $4/3$ and $n = p$ gives $p^2/(p^2-1)$.

Problem 10

Note. "Minimal" means: (a, b) is good, but neither $(a-b, b)$ nor $(a, b-a)$ is good (*excellent* pairs in the original).

Hint 1. $a\lceil b\nu \rceil - b\lfloor a\nu \rfloor$ is the determinant of $U = (a, \lfloor a\nu \rfloor)$ (just below the line $y = \nu x$) and $W = (b, \lceil b\nu \rceil)$ (just above it). Note that (m, m) is good.

Hint 2. Let $\lfloor (a+b)\nu \rfloor = \lfloor a\nu \rfloor + \lfloor b\nu \rfloor + e$ with $e \in \{0, 1\}$. Compute $f(a+b, b) = f(a, b) + b(1-e)$ and $f(a, a+b) = f(a, b) + ae$. So if (a, b) is good, *exactly one* of $(a+b, b)$, $(a, a+b)$ is good, and each good pair has at most one good "parent".

Hint 3. Hence the good pairs split into infinite chains, each starting at an excellent pair. For a good pair with $a > b$, $f(a-b, b) \in \{m, m-b\}$, and symmetrically for $b > a$. So an excellent pair has $\min(a, b) \leqslant m$. Moreover $f(a, b) \geqslant a(\lceil b\nu \rceil - b\nu)$ and $f(a, b) \geqslant b(a\nu - \lfloor a\nu \rfloor)$, so the other coordinate is bounded too, and there are finitely many excellent pairs.

Hint 4. Count excellent pairs according to $g = \gcd(a, b)$, which must divide m. Show that for each $g \mid m$ there are exactly m/g excellent pairs with $\gcd(a, b) = g$. Answer: $\sigma(m) = \sum_{d \mid m} d$ excellent pairs.